

Jahangirnagar University



Project Proposal On

Online Hall Management System for A F M Kamaluddin Hall

Project Title: Research Project Proposal

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Motivation:

Managing university residential halls with traditional paper registers is very slow and causes many clerical errors [1]. It is difficult for the administration to track student records, assign room seats accurately, and manage monthly fees by hand [1, 3]. Furthermore, physical ledgers offer no data protection, leaving sensitive student profiles vulnerable to loss or unauthorized access [3]. The lack of a digital system also creates communication gaps between students and hall authorities during daily operations [1].

This project is motivated by the urgent need to convert the manual workflows of A F M Kamaluddin Hall into a modern web platform [1, 4]. By using a secure web application, the hall can ensure error-free administration, role-based dashboard access, and transparent financial tracking [4]. Ultimately, this automation will establish a fast, safe, and highly efficient environment that protects student data and saves valuable time [1].

Objectives:

The primary goal of this project is to design, develop, and implement a secure, scalable, and user-friendly web application to automate the complete administrative and residential operations of A F M Kamaluddin Hall [1].

The specific objectives are:

- 1. Student Data Management:** To create a centralized digital system to easily store, update, and manage residential student records, eliminating paper errors [3].
- 2. Automated Seat Allocation:** To design an efficient sub-system for allocating and tracking hall room seats automatically with real-time accuracy [1].
- 3. Fee Collection & Auditing:** To implement a financial module for collecting yearly hall charges while keeping a transparent and clear transaction history [3, 4].
- 4. Administrative Controls:** To provide the admin with tools to manage staff accounts, update settings, and control user access permissions based on roles [4].
- 5. Advanced Web Security:** To protect student data and login credentials by using active session management, password encryption, and CSRF protection [4].
- 6. Responsive Interface:** To build a modern UI using Bootstrap that works smoothly on both desktops and mobile devices with easy password recovery options [1].

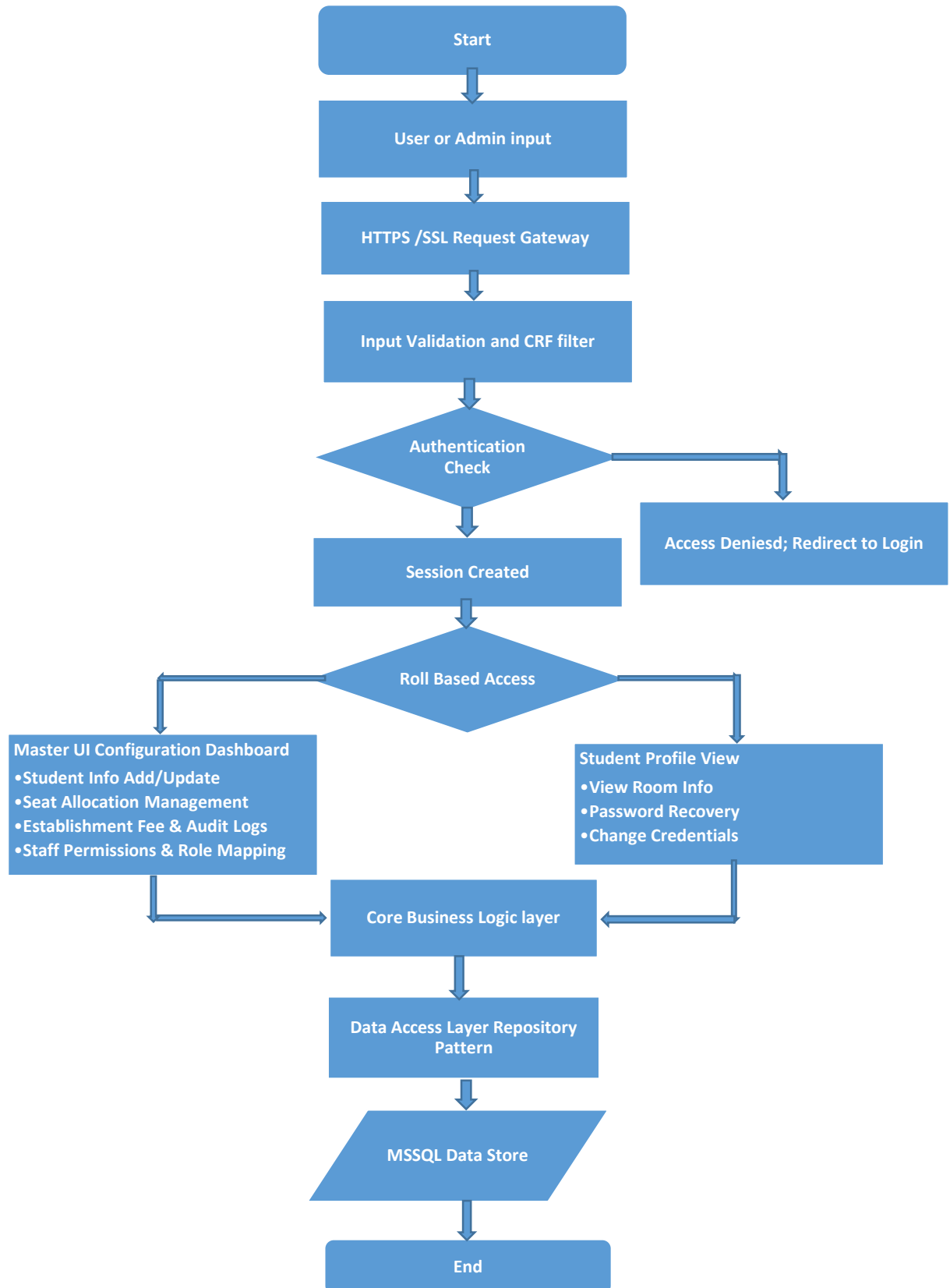
Introduction:

The A F M Kamaluddin Hall Management Web App is a modern web platform designed to automate all administrative and residential activities of the hall [1]. In traditional setups, managing hundreds of student files, room seats, and financial records involves heavy paperwork and frequent errors [2]. This centralized web system removes these inefficiencies by handling student profiles, room allocations, and fee histories in a single secure place [3].

The platform features a powerful Admin Control Panel for managing staff accounts and role-based user access rights, while allowing residential students to log in securely to track their individual room status and payment histories [4]. Developed using C#, ASP.NET Core MVC, Repository Pattern, and an MSSQL database, the system ensures high performance, scalability, and strong web protection [5].

Research Methodology & Flow of Work:

a. Workflow Diagram



Workflow Steps Defined:

- a. **Input & Validation:** Captures data via a responsive UI and instantly validates inputs to block malicious scripts [3].
- b. **Authentication & Role Checking:** Authenticates users, generates secure session tokens, and routes them to Admin, Staff, or Student panels based on roles [4].
- c. **Dashboard Execution & Persistence:** Processes dashboard actions through the Repository Pattern layer to execute secure transactions inside the MSSQL database [5].

Details of Work (Software Development):

This project follows standard software engineering steps to build a complete web app [1]:

- **Object-Oriented Analysis and Design:** The system architecture is modeled using entity objects representing Students, Staff, Seats, Rooms, and Invoices to keep code clean and manageable [2].
- **System Diagrams:** Formulates structural blueprints including Use Case Diagrams (user tasks), ER Diagram (MSSQL schema relationships), Class Diagram (C# data structures), and Data Flow Diagrams (data transit lines) [1, 3].
- **Layout & Dashboards:** Designs clean, Bootstrap-driven responsive interfaces for administrative entry logs, dynamic finance report generation, and real-time seat tracking metrics.
- **Coding & Enterprise Security:** Integrates strict enterprise-grade security controls directly within the source framework, covering secure user authentication, password hashing, active session controls, and explicit CSRF defense rules [4, 5].
- **Screenshots:** *(Functional software UI captures and layout modules will be placed here during the deployment phase).*

Discussion:

The deployment of this web application will make hall management exceptionally fast, automated, and paperless [1]. Developing with the ASP.NET Core framework and Repository Pattern ensures structural stability, easy code updates, and excellent data scalability under heavy transaction loads [2, 3].

However, ongoing server-side security monitoring is required to mitigate active network threats [4]. Future upgrades can integrate automatic Email/SMS transaction reminders and a native mobile application to further improve accessibility for the students of **A F M Kamaluddin Hall** [5].

Reference List to be used during working in your project:

1. Pressman, R. S. (2019). *Software Engineering: A Practitioner's Approach*. McGraw-Hill.
2. Troelsen, A. (2022). *Pro C# 10 with .NET 6*. Apress.
3. Elmasri, R. (2017). *Fundamentals of Database Systems*. Pearson.
4. Fowler, M. (2012). *Patterns of Enterprise Application Architecture*. Addison-Wesley.
5. Freeman, A. (2022). *Pro ASP.NET Core 6*. Apress.